

CH SHARIF AND SAEED HOSPITAL

Formula & Calculation Rule Book

Billing, Cash Settlement, Inventory, Attendance, Salary and Doctor Commission



Formula reference for development, QA and client sign-off.

Prepared by iSysware Software Solutions | Final Source of Truth v6.3

1. Formula governance

Formulas are templates plus staff/service/user-specific configuration. Where a client policy is not yet fixed, the system stores the method and values as configuration rather than hard-coding assumptions.

2. Service billing and discount formulas

Formula	Definition
Line Gross	Quantity x Unit Rate
Discount Amount - %	Line Gross x Discount %
Discount Amount - Fixed	Configured fixed discount, capped by allowed policy
Line Net	Line Gross - Discount Amount
Invoice Subtotal	Sum of all Line Net amounts + configured non-discountable charges
Amount Paid	Sum of successful payments - reversed/refunded payment amounts as applicable
Outstanding	Final Invoice Amount - Net Payments Applied

3. Panel/service discount

Formula	Definition
Panel Discount	Eligible Service Gross x Panel Service Discount %
Panel Service Net	Eligible Service Gross - Panel Discount
Patient/Panel Payable	According to panel contract split after discount; exact claim split is client-configurable

4. Doctor Commission formulas

Method	Formula
Percentage on Net - default	Commission = Net Eligible Service Amount x Commission %
Percentage on Gross	Commission = Service Gross x Commission %
Fixed Per Service	Commission = Fixed Amount x Eligible Completed Qty
Generated Commission	Sum eligible service commissions + approved adjustments - commission reversals
Commission Outstanding	Approved Generated Commission - Commission Payments Applied

Commission is based on completed eligible services and remains separate from Salary. Refund/cancel/reversal must reverse or adjust the related commission; never silently delete it.

5. Hospital share after discount/commission

Formula	Definition
Default Hospital Share	Net Service Amount - Doctor Commission
Example	5,000 gross - 500 discount = 4,500 net; 20% commission on net = 900; Hospital share = 3,600

6. Appointment / Admission payments

Formula	Definition
Appointment Remaining	Final Encounter Invoice - Appointment Advance Applied
Admission Hospital Due	Final Hospital Bill - all Hospital payments applied by Billing
Excess Advance	$\max(0, \text{Hospital Payments Applied} - \text{Final Hospital Bill})$
Pharmacy Due	Separate Pharmacy Invoice - Pharmacy Payments Applied

7. Attendance time formulas

Metric	Formula
Late Minutes	$\max(0, \text{Actual In} - (\text{Shift Start} + \text{Grace Minutes}))$
Early Exit Minutes	$\max(0, (\text{Shift End} - \text{Allowed Early Tolerance}) - \text{Actual Out})$
Worked Minutes	Actual Out - Actual In - Approved Break Minutes, with overnight shift normalized across date boundary
Present/Absent	Determined by configured minimum work/punch policy, approved leave and shift schedule

8. Salary formulas

Metric	Formula
Monthly Per-Day Rate	Monthly Base Salary / Configured Payroll Divisor
Period Base - monthly staff	Monthly Base x Eligible Period Fraction according to configured policy
Period Base - per day	Approved Payable Days x Per-Day Salary Rate
Absent Deduction	Absent Pay Units x Configured Per-Unit Deduction (often per-day rate)
Late Deduction	Configured staff rule: Fixed Per Event OR Late Minutes x Per-Minute Rate OR Threshold/Slab result
Early Exit Deduction	Configured staff rule: Fixed Per Event OR Early Minutes x Per-Minute Rate OR Threshold/Slab result
Generated Salary	Period Base + Salary Allowances + Approved Additions - Attendance Deductions - Other Approved Deductions +/- Adjustments
Salary Outstanding	Approved Generated Salary - Salary Payments Applied

Doctor Commission is NOT added into Generated Salary. It has its own generation, approval, payment and outstanding balance.

9. Billing cashier Balance Sheet

Metric	Formula
Expected Physical Cash Before Settlement	Opening Float + Cash Collections + Cash Received From Management - Cash Refunds - Authorized Cash Expenses
Online/Card/Bank Total	Sum of successful non-cash collections; report separately, do not add to physical cash
Cash Variance	Physical Cash Count - Expected Physical Cash Before Settlement
Closing Cash After Handover	Physical Cash Count - Cash Handed Over/Settled

Metric	Formula
Settlement Outstanding	Expected Handover Amount - Accepted Handover Amount, if partial settlement is allowed

10. Inventory user cash formulas

Metric	Formula
Available User Cash	Opening/Carry-forward Cash + Petty Cash/Advances Issued
Expected Remaining Cash	Available User Cash - Cash Purchases - Approved Cash Expenses + Cash Purchase Returns/Recoveries
Inventory Cash Variance	Physical Cash Count - Expected Remaining Cash
Closing User Cash	Physical Cash - Amount Handed Over in Settlement

11. Pharmacy cashier cash formulas

Metric	Formula
Expected Pharmacy Cash	Opening Float + Cash Pharmacy Sales + Cash Received - Cash Refunds - Authorized Cash Expenses
Pharmacy Cash Variance	Physical Cash - Expected Pharmacy Cash
Closing Pharmacy Cash	Physical Cash - Accepted Settlement Handover

12. General Inventory Stock Ledger formulas

Metric	Formula
Closing Quantity	Opening Qty + Purchases/Receipts + Department Returns + Transfer In + Positive Adjustments - Department Issues - Supplier Returns - Transfer Out - Negative Adjustments
Supplier Outstanding	Opening Supplier Outstanding + Credit Purchases - Supplier Payments - Purchase Returns - Approved Supplier Credits
Reorder Need - simple	If Available Qty <= Reorder Level, Suggested Qty = Target/Max Level - Available Qty (when target/max is configured)

13. Pharmacy stock formulas

Metric	Formula
Batch Closing Qty	Opening Batch Qty + Batch Receipts + Returns In + Transfers In - Dispensed/Sold - Supplier Returns - Transfers Out - Expiry/Damage/Negative Adjustments
Near Expiry	Expiry Date - Current Date <= Configured Near-Expiry Days AND Batch Qty > 0
FEFO Allocation	Choose valid non-expired available batch with earliest expiry first, subject to clinical/operational constraints
Pharmacy Supplier Outstanding	Opening + Credit Purchases - Supplier Payments - Supplier Returns - Approved Credits

14. Management consolidation formulas

Metric	Formula
Total Cash Held By Users	Sum of each active cash-holder Closing/Current Physical Cash
Total Pending Settlements	Sum of submitted but not fully accepted settlement amounts
Total Non-Cash Collections	Sum of successful Card/Bank/Online collections by selected period
User Accountability Delta	User Expected Cash - Accepted/Recorded Cash position, investigated via settlement variance and transaction drill-down

15. Rounding and audit rules

- Currency precision and rounding method are system settings; use the same method in UI, PDF, Excel and stored transaction snapshot.
- Every generated payroll/commission statement snapshots the calculation inputs/rules used at generation time so later Staff Profile changes do not rewrite history.
- Every invoice snapshots the rate/discount used; later Service Rate changes do not rewrite historical invoices.
- Every stock transaction is a movement with reference and actor; never recalculate old stock movement from today's master data.